Roll Number:

CS4.301: Data and Applications (Monsoon 2024)

Makeup Quiz

Maximum Marks: 15, Time: 40 minutes

- **Keep answers concise. State all assumptions.**
- **All questions are compulsory.**

Question 1

P				Q				R	
X	**Y**	**Z**		**X**	**Y**	**T**		**Y**	**V**
X1	Y1	Z1		X2	Y1	5		Y1	V1
X1	Y1	Z2		X1	Y2	3		Y3	V2
X2	Y2	Z2		X1	Y1	2		Y2	V3
X2	Y4	Z4		X3	Y3	8		Y2	V2

$$\pi_X\left(\sigma_{(P.Y = R.Y \wedge R.V = V2)}(P \times R)\right) - \pi_X\left(\sigma_{(Q.Y = R.Y \wedge Q.T < 8)}(Q \times R)\right)$$

How many tuples will be returned? Show working.

(3 marks)

~~PXR~~ $\sigma(\ldots)(P \times R)$

P.X	P.Y	P.Z	R.Y	R.V
X_2	Y_2	Z_2	Y_2	V_2

$\pi_x(\ ) \longrightarrow X_2$ <u>1 mark</u>

$\sigma(\ldots)(Q \times R)$

X_2	Y_1	5	Y_1	V_1
X_1	Y_2	3	Y_2	V_3
X_1	Y_1	2	Y_1	V_1

$\pi_x(\ ) \longrightarrow \{X_2, X_1\}$ or (X_2, X_1, X_1)

<u>1 mark</u>

$\{X_2\} - \{X_2 - X_1\} = \phi$

<u>1 mark</u>

Question 2

Consider a relation *R(A, B, C, D, E)* **that satisfies the following functional dependencies:** *{ABC → D, E → B, AD → C}*. **Decompose the schema in BCNF.**

(2 marks)

Answer :

R1(B,E) , R2 (A,C,D) , R4 (A,D,E) 1 mark

1 mark for correct set of steps

<u>Answer</u> (Show the steps leading to the BCNF decomposition):

Solution: There are two solutions:

Solution 1:

Table	$X^+ = ?$	New table 1	New table 2
$R(A, B, C, D, E)$	$ABC+ = ABCD$	$R_1(A, B, C, D)$	$R_2(A, B, C, E)$
$R_1(A, B, C, D)$	$AD+ = ACD$	$R_3(A, C, D)$	$R_4(A, B, D)$
$R_2(A, B, C, E)$	$E+ = BE$	$R_5(B, E)$	$R_6(A, C, E)$

Answer: $R_3(A, C, D), R_4(A, B, D), R_5(B, E), R_6(A, C, E)$.

Solution 2:

Table	$X^+ = ?$	New table 1	New table 2
$R(A, B, C, D, E)$	$E+ = BE$	$R_1(B, E)$	$R_2(A, C, D, E)$
$R_2(A, C, D, E)$	$AD+ = ACD$	$R_3(A, C, D)$	$R_4(A, D, E)$

Answer: $R_1(B, E), R_3(A, C, D), R_4(A, D, E)$.

Question 3

Let a prime attribute be one that appears in at least one candidate key. Let α and β be sets of attributes such that $\alpha \rightarrow \beta$ holds, but $\beta \rightarrow \alpha$ does not hold. Let A be an attribute that is not in α, is not in β, and for which $\beta \rightarrow A$ holds. We say that A is transitively dependent on α. We can restate our definition of 3NF as follows: *A relation schema R is in 3NF with respect to a set F of functional dependencies if there are no nonprime attributes A in R for which A is transitively dependent on a key for R.* **Show that this new definition is equivalent to the original one.**

(4 marks)

Suppose R is in 3NF according to the textbook definition. We show that it is in 3NF according to the definition in the exercise. Let A be a nonprime attribute in R that is transitively dependent on a key α for R. Then there exists $\beta \subseteq R$ such that $\beta \rightarrow A$, $\alpha \rightarrow \beta$, $A \notin \alpha, A \notin \beta$, and $\beta \rightarrow \alpha$ does not hold. But then $\beta \rightarrow A$ violates the textbook definition of 3NF since

- $A \notin \beta$ implies $\beta \rightarrow A$ is nontrivial.
- Since $\beta \rightarrow \alpha$ does not hold, β is not a superkey.
- A is not in any candidate key, since A is nonprime.

Now we show that if R is in 3NF according to the exercise definition, it is in 3NF according to the textbook definition. Suppose R is not in 3NF according to the textbook definition. Then there is a functional dependency $\alpha \rightarrow \beta$ that fails all three conditions. Thus,

- $\alpha \rightarrow \beta$ is nontrivial.
- α is not a superkey for R.
- Some A in $\beta - \alpha$ is not in any candidate key.

This implies that A is nonprime and $\alpha \rightarrow A$. Let γ be a candidate key for R. Then $\gamma \rightarrow \alpha, \alpha \rightarrow \gamma$ does not hold (since α is not a superkey), $A \notin \alpha$, and $A \notin \gamma$ (since A is nonprime). Thus A is transitively dependent on γ, violating the exercise definition.

1.5 mark for correct textbook/general definition of a table being in 3NF form

1 mark for highlighting the new definition in formal mathematical notation

1.5 mark for proving the transitive dependency

Not a formal proof is required , explanation based proof also works
Note : The restated definition is the one defined by Codd and is known as 'Codd's definition' for 3NF

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14.4.2 General Definition of Third Normal Form

Definition. A relation schema R is in **third normal form (3NF)** if, whenever a *nontrivial* functional dependency $X \rightarrow A$ holds in R, either (a) X is a superkey of R, or (b) A is a prime attribute of R.[13]

Question 4

SQL provides an *n*-ary operation called coalesce**, which is defined as follows:**
coalesce($A_1, A_2, \ldots, A_n$) **returns the first nonnull** A_i **in the list** $A_1, A_2, \ldots, A_n$**, and returns** *null* **if all of** $A_1, A_2, \ldots, A_n$ **are** *null***.**

Let *a* **and** *b* **be relations with the schemas** *A(name, address, title)*, **and** *B(name, address, salary)*, **respectively. Show how to express** *a* natural full outer join *b* **using the** full outer-join **operation with an** on **condition and the** coalesce **operation. Make sure that the result relation does not contain two copies of the attributes** *name* **and** *address*, **and that the solution is correct even if some tuples in** *a* **and** *b* **have null values for attributes** *name* **or** *address*.

(4 marks)

```sql
SELECT COALESCE(a.name, b.name) AS name,
    COALESCE(a.address, b.address) AS address,
    a.title,
    b.salary
FROM a FULL OUTER JOIN b
    ON a.name = b.name AND
    a.address = b.address;
```

1.5 mark for highlighting the idea /explanation ,

2 marks for the correct query (partial marks based on combination of explanation and written query)

0.5 marks for correct join conditions

It's an easy question if you get the definition right

Note: 4 for correct query and a detailed answer/ thought process of getting to the answer

Max 1 for similar idea

Question 5

Rewrite the following query without using the with construct. The query must run; no partial marks will be awarded.

with *dept_total (dept_name, value)* **as**
(**select** *dept_name*, **sum***(salary)*
from *instructor*
group by *dept_name)*,
dept_total_avg (value) **as**
(**select avg** (*value*)
from *dept_total*)
select *dept_name*
from *dept_total*, *dept_total_avg*
where *dept_total.value* >= *dept_total_avg.value*;

(2 marks)

SELECT dept_name
FROM (SELECT dept_name, SUM(salary) AS value FROM instructor GROUP BY dept_name) AS dept_total,
(SELECT AVG(value) AS value FROM (SELECT dept_name, SUM(salary) AS value FROM instructor GROUP BY dept_name) AS x) AS dept_total_avg
WHERE dept_total.value >= dept_total_avg.value

2 marks for correct query (No partial marking)